Granularity is ubiquitous in nature: light is made of photons, the particles of light. The energy of electrons in atoms can acquire only certain values and not others. The purest air is granular, and so, too, is the densest matter. Once it is understood that Newton's space and time are physical entities like all others, it is natural to suppose that they are also granular. Theory confirms this idea: loop quantum gravity predicts that elementary temporal leaps are small, but finite.

The idea that time could be granular, that there could be minimal intervals of time, is not new. It was defended in the seventh century by Isidore of Seville in his *Etymologiae*, and in the following century by the Venerable Bede in a work suggestively entitled *De Divisionibus Temporum* ("On the Divisions of Time"). In the thirteenth century, the great philosopher Maimonides writes: "Time is composed of atoms, that is to say of many parts that cannot be further subdivided, on account of their short duration."[53] The idea probably dates back even further: the loss of the original texts of Democritus prevents us from knowing whether it was already present in classical Greek atomism.[54] Abstract thought can anticipate by centuries hypotheses that find a use—or confirmation—in scientific inquiry.

The spatial sister of *Planck time* is *Planck length*: the minimum limit below which the notion of length becomes meaningless. Planck length is around 10^{-33} centimeters: a millionth of a billionth of a billionth of a billionth of a millimeter. As a young man at university, I fell in love with the problem of what happens at this extremely small scale. I painted a large sheet with, at its center, in red, a glittering:



I hung it up in my bedroom in Bologna and decided that my goal would be to try to understand what happens down there, at the very tiny scale where time and space cease to be what they are—all the way to the elementary